EXAMPLE 4.8

A conducting rod of length 0.5 m falls freely from the top of a building of height 7.2 m at a place in Chennai where the horizontal component of Earth's magnetic field is 4.04×10^{-5} T. If the length of the rod is perpendicular to Earth's horizontal magnetic field, find the emf induced across the conductor when the rod is about to touch the ground. (Assume that the rod falls down with constant acceleration of 10 m s^{-2})

Solution

$l = 0.5$ m; $h = 7.2$ m; $u = 0$ m s^{-1};
$g = 10$ m s^{-2}; $B_H = 4.04 \times 10^{-5}$ T

The final velocity of the rod is

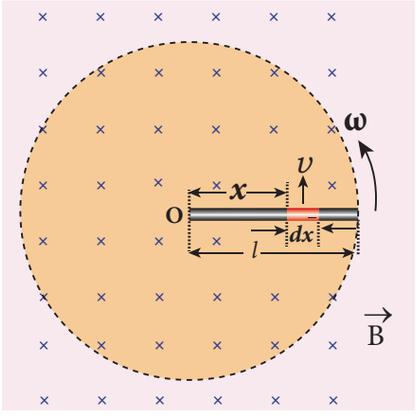
$$v^2 = u^2 + 2gh = 0 + (2 \times 10 \times 7.2) = 144$$
$$v = 12 \text{ m s}^{-1}$$

The magnitude of the induced emf when the rod is about to touch the ground is

$$\varepsilon = B_H l v = 4.04 \times 10^{-5} \times 0.5 \times 12$$
$$= 242.4 \ \mu\text{V}$$

EXAMPLE 4.9

A copper rod of length l rotates about one of its ends with an angular velocity ω in a magnetic field B as shown in the figure. The plane of rotation is perpendicular to the field. Find the emf induced between the two ends of the rod.



Solution

Consider a small element of length dx at a distance x from the centre of the circle described by the rod. As this element moves perpendicular to the field with a linear velocity $v = x\omega$, the emf developed in the element dx is

$$d\varepsilon = Bv\,dx \quad = B(x\omega)dx$$

This rod is made up of many such elements, moving perpendicular to the field. The emf developed across two ends is

$$\varepsilon = \int d\varepsilon = \int_0^l B\omega x\,dx = B\omega\left[\frac{x^2}{2}\right]_0^l$$
$$\varepsilon = \frac{1}{2}B\omega l^2$$

4.2 EDDY CURRENTS

According to Faraday's law of electromagnetic induction, an emf is induced in a conductor when the magnetic flux passing through it changes. However, the conductor need not be in the form of a wire or coil.

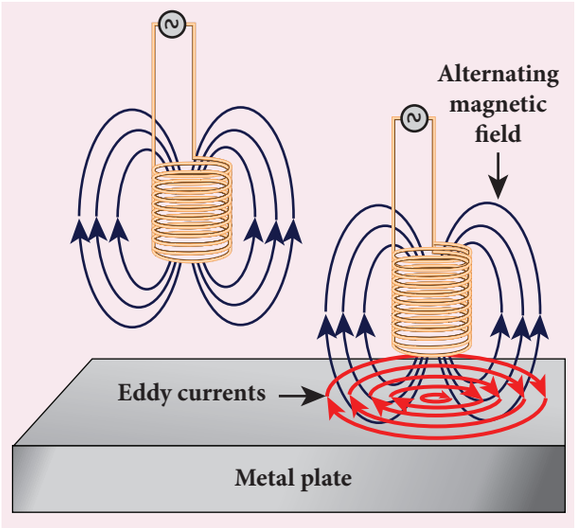


Figure 4.10 Eddy currents

12th Physics_Vol-1_Unit 4_English Medium.indd 207 23-12-2022 17:56:49